

Shift-Register Circuit

Given $g(x)$, $\deg(g(x)) = n-k$

E.g. $g(x) = x^3 + x + 1$
 $g_3=1 \quad g_2=0 \quad g_1=1 \quad g_0=1$

- ① # registers = $\deg(g(x)) = n-k$
 $=$ # check bits
 (with initial value of 0)

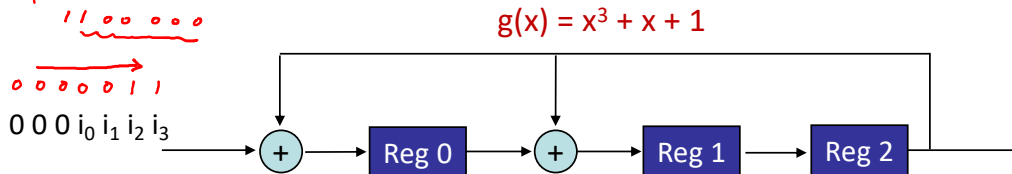


- ② Coefficients of $g(x)$
 $\Rightarrow$ how to connect
 how to add XOR gates
- ③ Input from left
 highest-degree term coefficient of $a(x)$ first
- ④ after n clock ticks,
 $(n-k)$ check bits in the registers
 $r(x)$

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Shift-Register Circuit Implementation

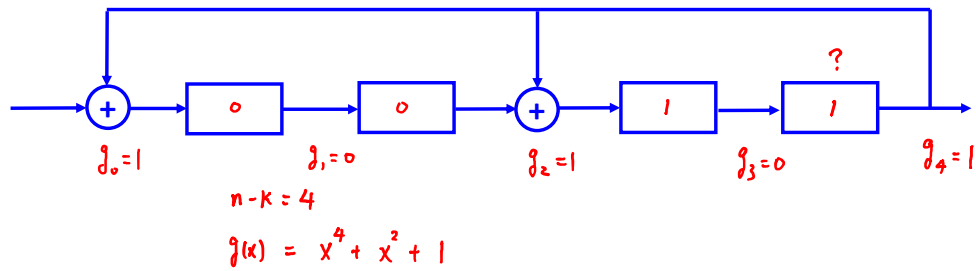
4 info bits: (1100)



Clock	Input	Reg 0	Reg 1	Reg 2
0	-	0	0	0
→ 1	<u>1</u> = i_3	$1+0 = 1$	$0+0 = 0$	$0 = 0$
2	$1 = i_2$	1	1	0
3	$0 = i_1$	0	1	1
4	$0 = i_0$	<u>1</u>	<u>1</u>	<u>1</u>
→ 5	<u>0</u>	$0+1 = 1$	$1+1 = 0$	$1 = 1$
6	0	1	0	0
7	0	0	1	0

Check bits: $r_0 = 0$ $r_1 = 1$ $r_2 = 0$
 $\Rightarrow r(x) = x$

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E.g. info bits (1 1 1 1) $\Rightarrow$ codeword: (1 1 1 1 1 0 0)

$$\begin{array}{r}
 10101 \overline{) 11110000} \\
 \underline{10101} \\
 10110 \\
 \underline{10101} \\
 1100
 \end{array}$$

$$r(x) = x^3 + x^2$$

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$$g(x) = x^4 + x^2 + 1 \quad n-k=4, \quad k=4, \quad n=8$$

* Example undetectable $L=8$?

$$e = [1 \, ? \, ? \, ? \, ? \, ? \, ? \, 1]_{1 \times 8} \quad e(x) = g(x) \cdot c(x)$$

$$e(x) = x^7 + ??? + 1 = (x^4 + x^2 + 1) \cdot c(x)$$

$$\Rightarrow c(x) = x^3 + ?x^2 + ?x + 1$$

$$\text{let's pick } c(x) = x^3 + 1 \Rightarrow e(x) = g(x) \cdot c(x) = x^7 + x^5 + x^4 + x^3 + x^2 + 1$$

$$\Rightarrow e = [1 \, 0 \, 1 \, 1 \, 1 \, 0 \, 1]_{1 \times 8}$$

$$L=8, \quad M=6$$

$$\begin{array}{r}
 e(x) \\
 11 \\
 g(x) 10101 \\
 + \\
 g(x) \cdot x^3 10101000 \\
 \hline
 e = [10111101]_{1 \times 8}
 \end{array}$$

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* Example undetectable $M = 2$? (2-bit error)

$$e = [0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1]_{1 \times 8}$$

$$g(x) = x^4 + x^2 + 1$$

$$e = [1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0]_{1 \times 8}$$

$$1 \ 0 \ 1 \ 0 \ 1$$

$$\Rightarrow e(x) = g(x) \cdot x \cdot (x^2 + 1)$$

$$\begin{array}{rcl} 1 \ 0 \ 1 \ 0 \ 1 & \leftarrow & g(x) \\ 1 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 & \leftarrow & g(x) \cdot x^2 \\ \hline 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 & \leftarrow & g(x) \cdot (x^2 + 1) = e(x) \end{array}$$

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Standard Generator Polynomials



Name	Polynomial	Used in
CRC-8-CCITT	$x^8 + x^2 + x + 1$	ATM header
CRC-10	$x^{10} + x^9 + x^5 + x^4 + x + 1$	ATM
CRC-12	$x^{12} + x^{11} + x^3 + x^2 + x + 1$	Bisync
CRC-16-CCITT	$x^{16} + x^{12} + x^5 + 1$	HDLC
CRC-32 <u> </u>	$x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1$	IEEE 802

$$FUE < \frac{1}{2^{n-k}} = \frac{1}{2^{32}} \approx 2.33 \cdot 10^{-10}$$

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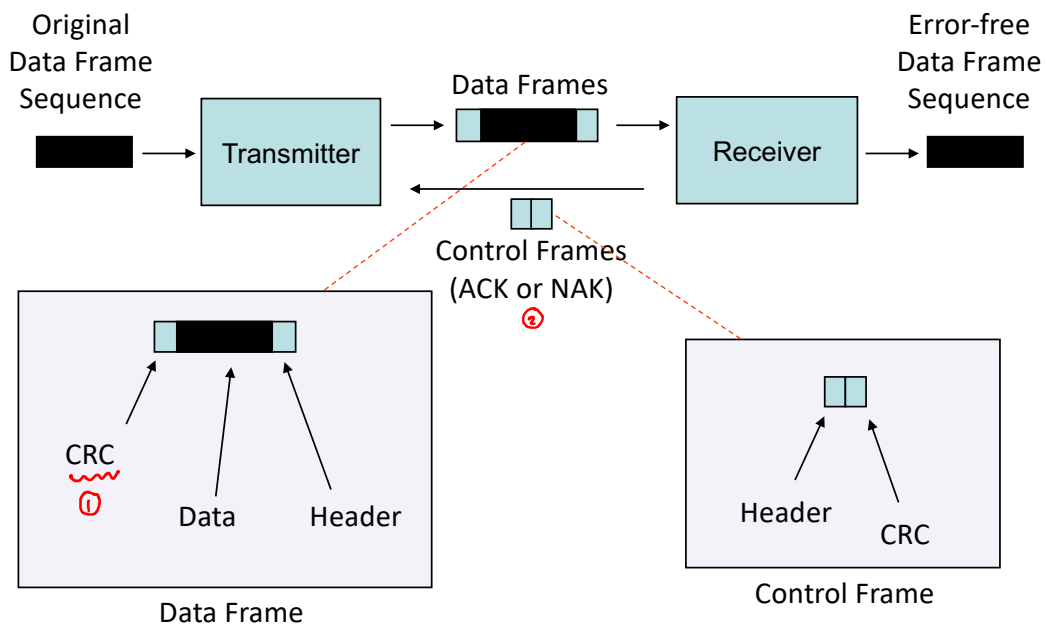
ARQ (Automatic Repeat reQuest)

⊕ Three Design Goals of ARQ Protocols

- ➡ **Goal #1:** to ensure that each data frame is delivered **error-free**
- ➡ **Goal #2:** to ensure that each data frame is delivered **exactly once without duplication**
- ➡ **Goal #3:** to ensure that data frames are delivered **in order**

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ARQ Basics



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5 Basic Elements:

- ① Error Detection (CRC)
- ② Control Frames : ACK , NAK
Required optional
- ③ Retransmission if needed
- ④ Timeout
- ⑤ Sequence Number (SN)

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SN:

- SN of data frame : position in the sequence
- SN carried in control frames :
regarding whether or which data frames in the sequence
have been received so far
"cumulative acknowledgment"

E.g. $\text{ACK}(20)$ $\Rightarrow$ all data frames with $\text{SN} < 20$
have been received correctly
 $\Rightarrow$ the next data frame that receiver
is waiting for is #20

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